Classical Mechanics

Labix

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Contents

1 Newtonian Mechanics

1.1 Galilean Space and Galilean Transformations

We begin with an experimental fact in classical mechanics. The experimental facts should be thought of as a set of axioms for classical mechanics to operate on. The question of accuracy of these axioms modeling real life phenomena is a different question.

Axiom 1.1.1 (Space and Time)
Our space is a three dimensional Euclidean space and time is one dimensional.

Definition 1.1.2 (The Galilean Spacetime Structure)
A Galilean spacetime structure consists of the following data.

- An affine space $(A, \mathbb{R}^4, +)$. The vector space $\mathbb{R}^4$ are called the parallel displacements of A.
- A metric $d : \mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}$ on the parallel displacements of A defined by

$$d((t_1, x_1, x_2, x_3), (t_2, y_1, y_2, y_3)) = \begin{cases} |t_2 - t_1| & \text{if } t_1 \neq t_2 \\ \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + (y_3 - x_3)^2} & \text{otherwise} \end{cases}$$

In this case, we call A a Galilean space.

The definition of Galilean spacetime structure incorporates our first axiom in its fullest.

Notice that we could have equally defined a Galilean spacetime structure by a non-zero linear map $t : \mathbb{R}^4 \to \mathbb{R}$ together with the standard Euclidean metric on $\ker(t)$. Then up to isomorphism, the non-zero linear map $t : \mathbb{R}^4 \to \mathbb{R}$ is just the projection to the first coordinate and the metric structure on $\ker(t)$ can be extended to any affine subspace $x + \ker(t)$ for any $x \in \mathbb{R}^4$.

Definition 1.1.3 (Galilean Transformation)
Let A be a Galilean space. Let $f : A \to A$ be an affine transformation. We say that f is the induced map of vector spaces $\overline{f} : \mathbb{R}^4 \to \mathbb{R}^4$ is a distance preserving function of metric spaces.

Definition 1.1.4 (Types of Galilean Transformation)
Let A be a Galilean space. Let $g : A \to A$ be a Galilean transformation and $\overline{g} : \mathbb{R}^4 \to \mathbb{R}^4$ the induced map of vector spaces and metric spaces.

- We say that g is a uniform motion with velocity $v \in \mathbb{R}^3$ if
$$\overline{g}(t, x) = (t, x + vt)$$
- We say that g is a translation if there exists $s \in \mathbb{R}$ and $a \in \mathbb{R}^3$ such that
$$\overline{g}(t, x) = (t + s, x + a)$$
- We say that g is a rotation if there exists an orthogonal map $G : \mathbb{R}^3 \to \mathbb{R}^3$ such that
$$\overline{g}(t, x) = (t, G(x))$$

Definition 1.1.5 (The Galilean Group)
Let A be a Galilean space. Define the Galilean group of A to be

$$\text{Gal}(A) = \{f : A \to A \mid f \text{ is a Galilean transformation }\}$$

Proposition 1.1.6
Let A be a Galilean space. Then every Galilean transformation can be uniquely decomposed as the composition of a rotation, translation and uniform motion.

Proposition 1.1.7
Let A be a Galilean space. Then $\text{Gal}(A)$ is a Lie group of dimension 10.

1.2 Galilean Coordinate Systems

Definition 1.2.1 (The Standard Galilean Space)
Define the standard Galilean space to be $\mathbb{R}^4$ equipped with the action $\mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$ that is vector addition.

Lemma 1.2.2
Let A be a Galilean space. Then there exists a bijective Galilean transformation $f : A \to \mathbb{R}^4$.

Definition 1.2.3 (Galilean Coordinate Systems)
Let A be a Galilean space. A Galilean coordinate system is a bijective Galilean transformation $A \to \mathbb{R}^4$.

Thus giving a bijective Galilean transformation is the same the act of giving concrete coordinates to a Galilean space. It is commonly called in literature as frames of references.

Definition 1.2.4 (Uniform Movement)
Let A be a Galilean space. Let $\varphi_1, \varphi_2 : A \to \mathbb{R}^4$ be two Galilean coordinate systems of A. We say that φ_2 moves uniformly with φ_1 if

$$\varphi_1 \circ \varphi_2^{-1} : \mathbb{R}^4 \to \mathbb{R}^4$$

is a Galilean transformation.

Axiom 1.2.5 (Galileo's Principle of Relativity)
Let A be a Galilean space. Then there exists a collection of coordinate system $S = \{\varphi_i : A \to \mathbb{R}^4 \mid i \in I\}$ such that the following are true.

- Any law of nature holds true in $\text{im}(\varphi_i)$ for any $i \in I$.
- If $f : A \to \mathbb{R}^4$ is a Galilean coordinate system that moves uniformly with φ for some $\varphi \in S$, then $f \in S$.

Any Galilean coordinate system in such a collection is called an inertial coordinate system.

Notice that for instance, if φ is an inertial frame of reference and $g : \mathbb{R}^4 \to \mathbb{R}^4$ is a Galilean transformation, then $\varphi \circ g$ is also an inertial frame of reference.

1.3 Motions

Definition 1.3.1 (Motions)
A motion in $\mathbb{R}^3$ is a C^∞ function $x : [a, b] \to \mathbb{R}^3$ for some $a, b \in \mathbb{R}$.

Let X, Y be sets. Let $f : X \to Y$ be a function of sets. Recall that the graph of f is the subset of $X \times Y$ defined by

$$\Gamma(f) = \{(x, f(x)) \in X \times Y \mid x \in X\}$$

Definition 1.3.2 (World Lines)
Let A be a Galilean space. A world line of A is the preimage of a graph of a motion $x : \mathbb{R} \to \mathbb{R}^3$ under a choice of Galilean coordinate system $\varphi : A \to \mathbb{R}^4$.

Definition 1.3.3 (Mechanical Systems)
Let A be a Galilean space. Let $n \in \mathbb{N} \setminus \{0\}$. A mechanical system of n points in A is a collection of C^∞ function $x_1, \dots, x_n : \mathbb{R} \to \mathbb{R}^3$ together with a choice of Galilean coordinate system $\varphi : A \to \mathbb{R}^4$.

A mechanical system is closed when we have included all bodies whose interactions play a role in the study of the given phenomena. Normally, this would mean that the universe is the unique closed system we have to consider for any phenomena, but because most of the bodies contribute negligibly, we allow for a more lax definition of closed systems when implementing for particular phenomena.

Axiom 1.3.4 (Newton's Principle of Determinacy)
Let A be a Galilean space. Let $n \in \mathbb{N} \setminus \{0\}$. Let $x : \mathbb{R} \to \mathbb{R}^{3n}$ be a mechanical system. Then there exists a smooth function $F : \mathbb{R}^{3n} \times \mathbb{R}^{3n} \times \mathbb{R} \to \mathbb{R}^{3n}$ such that

$$\left(\frac{d^2x_1}{dt^2}, \dots, \frac{d^2x_n}{dt^2}\right) = F\left(x_1, \dots, x_n, \frac{dx_1}{dt}, \dots, \frac{dx_n}{dt}, t\right)$$

Equivalently, for each $1 \leq i \leq n$, there exists $F_i : \mathbb{R}^{3n} \times \mathbb{R}^{3n} \times \mathbb{R} \to \mathbb{R}^{3n}$ such that

$$\frac{d^2x_i}{dt^2} = F_i\left(x_1, \dots, x_n, \frac{dx_1}{dt}, \dots, \frac{dx_n}{dt}, t\right)$$

Proposition 1.3.5
Let A be a Galilean space. Let $n \in \mathbb{N} \setminus \{0\}$. Let $x : \mathbb{R} \to \mathbb{R}^{3n}$ be a closed mechanical system. Let F be the smooth function from Newton's principle of determinacy for the system x. Then the following are true.
- F is not a function of time t.
- If $G : \mathbb{R}^3 \to \mathbb{R}^3$ is an orthogonal linear map, then $F_i(G \circ x_1, \dots, G \circ x_n, \dots, G \circ \frac{dx_1}{dt}, \dots, G \circ \frac{dx_n}{dt}) = G \circ F_i(x_1, \dots, x_n, \frac{dx_1}{dt}, \dots, \frac{dx_n}{dt})$
- For $1 \leq i \leq n$, F_i is a function of $x_j - x_k$ and $\frac{dx_j}{dt} - \frac{dx_k}{dt}$ for $1 \leq j < k \leq n$.